PEDIATRIC PROTOCOLS

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Emergency Phone-Line to Vanderbilt Children's ER
Comprehensive Regional Pediatric Center (CRPC) -
(615) 936-7575

GENERAL PEDIATRIC CARE GUIDELINES

The key to quality pediatric care lies in the realization that children are not small adults. Scaled down equipment and smaller drug dosages are only the beginning. Pediatrics requires a different approach to patient care. The following guidelines should be kept in mind when treating pediatrics:

- The age range in pediatrics can make obtaining a history difficult but you should never dismiss the child's history.
- Initiate assessment of the pediatric patient using the PAT (Pediatric Assessment Triangle) to quickly evaluate appearance, work of breathing, and circulation status. A rapid cardiopulmonary assessment should be performed on all patients on initial contact and after each intervention.
- Cardiac arrest is seldom a sudden event. It is most often the results of a progressive deterioration of the circulatory (shock) and respiratory (hypoxia) systems.
- Hypoxia produces a reflex bradycardia in children. Any change in respiratory rate should be evaluated for a corresponding change in heart rate and vice-versa.
- Aggressive airway control and ventilation should be a priority, however in cardiac arrest, high quality CPR and early defibrillation should never be delayed.
- Intraosseous (IO) lines are the preferred route of access for all cardiac arrest victims. IO's may also be placed in a pediatric patient that is critically ill and needs immediate life-saving intervention. For example, such a patient may be displaying signs and symptoms of inadequate tissue perfusion (pale, cool, cyanotic or diaphoretic skin), altered level consciousness (lethargy), or profound hypotension defined as a systolic blood pressure less than (70 + (age in years' x 2)).
- Any medication directed to be given IV may also be given IO.
- Try to involve the parents as much as possible without compromising patient care.
- In a case of obvious death, CPR should be performed if it is the parent's wishes. Never leave the parent with the impression that something else could have been done.
- The cardiac monitor and other necessary equipment needed for resuscitation should be <u>taken to the immediate side of any unconscious patient</u> (or any patient presumably or likely to be in cardiac arrest until proven otherwise.
- Capnometry (waveform capnography) should be used in any critical pediatric patient to confirm placement of advanced airways and to monitor the perfusion, ventilation, and respiration status of the patient (This includes side stream or mainstream sampling). Sumner EMS carries pediatric side stream (nasal cannula) and mainstream ETCO2 monitoring equipment.
- Carbon Dioxide (CO) levels should be assessed on any patient presenting with altered mental status where there is potential or suspicion for CO exposure.
- Parents/guardians should be allowed to ride with the pediatric patient to the hospital and attempts should be made to not separate children from parents, unless the child is critical and it is best for everyone for the parent to ride elsewhere. (Also see COVID protocol)
- Temperature assessment for pediatric patients should be done on patients presenting with general sickness, febrile illness, seizures, hypothermia, or heat related illness.
- Sumner County EMS ambulances are only equipped with temperature monitoring equipment to perform oral thermometry or if available, the Zoll monitor temperature skin contact probe may also be adhered to the patient. At no time will EMS personnel assess rectal temperatures in the pre-hospital environment.
- Do not use an adult (normal size) finger lancet on an infant or small child's finger. This can lead to a laceration that will require treatment itself. Obtain samples from the heel in pediatric patients this small.

PEDIATRIC – EMERGENCY NEEDLE CRICOTHYROTOMY

OVERVIEW:

A cricothyrotomy is a procedure to establish an emergency airway. It is an invasive procedure with multiple inherent complications and should be performed only on patients that are at high risk of death if an immediate airway is not established. One must first consider/attempt all alternative airway measures (e.g. OPA, NPA, ET intubation, supraglottic device, etc.). While every attempt should be made to transport to the closet emergency department for a more controlled setting, but no patient under the care of Sumner County EMS should die secondary to airway obstruction.

REQUIREMENTS:

- Be a licensed paramedic credentialed through the Deputy Chief of Training.
- Must have completed bi-annual training sessions as required by Sumner County EMS.
- No longer required to contact on-line medical control, this is a standing order now.

INDICATIONS:

- Emergent Need Only.
- **Inability to intubate and inability to ventilate.**
- Indicated for extremis presentations of**:**
    - Foreign body airway obstruction.
    - Airway burns.
    - Anatomical injury
    - Anaphylaxis unresolved by Epi
    - Epiglottitis/croup unresolved by Epi

COMPLICATIONS:

- Bleeding.
- Misplacement (esophageal or soft-tissue placement).
- Damage to surrounding structures such as vocal cords, esophageal or tracheal damage.
- Infection.

CONTRAINDICATIONS:

Given that you will only be performing this procedure on patients who have a very high probability of dying without it, most contra-indications would therefore be relative. The following are examples of situations that may prohibit needle cricothyrotomy:

- Age > 10 years old / Adult sized, refer to the adult cricothyrotomy protocol if needed.
- Able to ventilate with less invasive therapies.
- Any causes preventing identification of appropriate anatomical landmarks.

Continue on next page…

PEDIATRIC – EMERGENCY NEEDLE CRICOTHYROTOMY

EQUIPMENT NEEDED:

• Non latex gloves	• ½ inch medical tape
• Sharps container	• 3.0 ET tube (connector)
• Oxygen Supply BVM	• NS Flush
• ChloraPrep/antiseptic	• Neo-T Resuscitator Device
• 14G IV Angiocath	• 90-degree IV extension set

Find landmark, same approach as adult.

↓

Prep skin with antiseptic.

↓

Insert 14G cath at 45- degree angle with hub held caudally (toward the feet).

↓

Withdraw air during insertion through NS flush half filled with fluid.

↓

Assess for penetration into trachea by watching for air bubbles in syringe.

↓

Careful not to insert needle too far, the diameter of the pediatric trachea is similar in size to the child's pinky finger

↓

Advance the catheter flush to the skin, remove needle and dispose properly.

↓

Attach IV/IO extension set directly to 14G catheter.

↓

Attach 3.0 mm ETT hub to other end of IV/IO extension set.

↓

Attach Neo-T style resuscitator to ETT connector and initiate ventilations.

↓

Ensure that the Neo-T is set to "high" flow rate (red).

↓

Adjust built-in PEEP dial titrated to effect to ensure effective exhalation.

↓

Confirm that chest rise and fall is present.

↓

The O2 source powering the Neo-T resuscitator should be running at high-flow (15 lpm +).

↓

Secure as soon as possible with tape provided. This should be treated as fragile as a pediatric/infant intubation and the cannula should not be released until it can be secured.

FACILITATED INTUBATION AND RSI

OVERVIEW:

Rapid Sequence Intubation (RSI) is a series of maneuvers utilizing sedation and paralysis to establish an advanced airway in a critically ill patient. This is an advanced procedure with a potential for high-risk complications and should only be performed as an absolute life-saving procedure. It should only be performed after all other less invasive forms of airway control have been attempted or considered. At no time should a paramedic feel pressured to perform this procedure if he or she is not comfortable with its application on a given patient. ***Updated 2019 - All paramedics may initiate this procedure with the administration of choice sedatives (chemically facilitated intubation). However, you must meet the requirements below in order to perform full RSI, to include administration of paralytics.***

REQUIREMENTS TO PERFORM RSI:

- Be a licensed paramedic and employee of Sumner County EMS for at least 6 months, or experienced personnel may be approved by the Medical Director.
- Be in good standing with the service regarding clinical issues.
- Complete bi-annual airway, RSI and cricothyrotomy training courses.

INDICATIONS:

To establish an airway in a patient who is at risk of death secondary to loss of airway or inability to ventilate, and the airway cannot be controlled by conventional means. Examples of patients in which pre-hospital RSI might be indicated include, but are not limited to the following:

- Facial or head trauma patients with loss of airway control.
- Severe respiratory distress with hypoxia and/or respiratory exhaustion.
- Burn patients with airway involvement and respiratory distress.
- Overdose with loss of airway protection and hypoxia.

CONTRAINDICATIONS:

- Allergy to any one of the agents.

CONTRAINDICATIONS TO SUCCINYLCHOLINE:

- History of malignant hyperthermia.
- Renal failure.
- Spinal cord injury greater than 24 hours old or neuromuscular disease.
- Severe burns greater than 8 hours old.
- Massive crush injuries.
- Pesticide poisoning.
- Penetrating eye injuries.

Continue on next page…

FACILITATED INTUBATION AND RSI (Procedure)

Preparation:

- Assemble and check all needed equipment and medications and anticipate a difficult airway.

Preoxygenate:

- Allow patient to breathe high flow O2, ventilate only as needed to increase SpO2 (avoid gastric distention). Place on a nasal cannula @ 15 lpm and leave in place until the procedure is completed.

Pretreatment:

- Attach Capnography and begin monitoring early into the procedure.
- **In children less than 8 years old, *consider* administering 0.02 mg/kg of Atropine IV/IO.**

Give **SEDATIVE** (Induction) – Use appropriate, available induction agent:

- Ketamine 1-2 mg/kg IV/IO.
  - First choice induction agent (sedative).
  - Avoid if patient is hypertensive, tachyarrhythmia or acute MI.
- Etomidate 0.25 mg/kg IV/IO, when Ketamine is contraindicated **(first choice alternate to Ketamine)**.
- Versed 0.2 mg/kg, max of 5 mg IV/IO (only used as an alternate to Etomidate).

*****CONSIDER ATTEMPTING FACILITATED INTUBATION AFTER AVAILABLE INDUCTION AGENT*****

******REFER TO CONTRAINDICATIONS FOR SUCCINYLCHOLINE ON PAGE 6 PRIOR TO GIVING!!!!******

Give PARALYTIC (short-acting)

- Administer Succinylcholine 1.5 mg/kg IV/IO, maximum dose of 100 mg, or Rocuronium 1 mg/kg IV/IO, maximum dose of 100 mg.

Placement and Proof:

- Intubate when patient becomes flaccid, often after fasciculation's. If the patient cannot be intubated *after 2 attempts*, then use an alternative airway such as the iGel airway or basic airway adjuncts and continue to ventilate patient until the Succinylcholine wears off.
- Confirm placement with EtCO2 detector, Capnography/Capnometry, auscultation, etc. **(minimum of 5 verification techniques)**.

Procedure Continued on Next Page…

FACILITATED INTUBATION AND RSI (Continued) – POST-INTUBATION MANAGEMENT

For any patient who is already intubated (interfacility transfers), ALL PARAMEDICS can begin with this section to maintain sedation, go to long-acting paralytics as needed, see below…

Post-Intubation Management:

- 100% O2, and titrate to > 92% when possible.
- Maintain EtCO2 of 35-45 mmHg when possible.
- Secure ET Tube.
- DO NOT overinflate with the BVM for risk of causing barotrauma.
- Use a PEEP valve (2 - 5 cmH2O) in patients with pulses and a stable BP.
- **Place an orogastric tube as soon as time permits.**

Give maintenance **SEDATION** and **PAIN CONTROL**… This **MUST BE DONE**!

Ketamine – 0.5-1 mg/kg IV/IO, one time dose, if Ketamine was used for induction agent prior to RSI

Versed – 0.05-0.1 mg/kg IV/IO to a maximum of 5 mg:

- Repeat 0.05-0.1 mg/kg as necessary and titrate to desired effect.

Fentanyl – 1 mcg/kg IV/IO may also be given:

- Repeat as necessary and titrate to desired effect.

Administer long-acting paralytic as indicated after correct placement is assured:

- Rocuronium (Zemuron) 1 mg/kg, max of 100 mg IV/IO.
  - Actual dose range is 0.6-1 mg/kg. We can give 1 mg/kg for ease of administration.
- Norcuron (Vecuronium) 0.1 mg/kg, max of 10 mg IV/IO.
  - Second choice if Rocuronium is unavailable…

Transport without delay, as safely as possible…

DOCUMENT WELL… Include in your documentation the reason the procedure was required, the procedure used, intubation verification **(5)** methods, **EtCO2 must be used and documented**, and the patient's response.

Guidelines for Cuffed ETT Utilization in the Pediatric Patient

(Adopted from Tennessee EMS for Children's Education Guidelines)

When using a cuffed ETT for a pediatric patient:

1. Select the appropriate size ETT (typically ½ size smaller than the recommended uncuffed ETT size as per length-based resuscitation tape).

2. Check the integrity of the cuff prior to insertion.

3. After insertion of the ETT, inflate the cuff as necessary to achieve minimal air leak around the ETT (amount of air not to exceed the manufacturer's specification for maximum air inflation).

4. Completely deflate the cuff prior to removal of the ETT.

It is accepted practice to use cuffed ETTs in any patient. However, providers must understand that they may not need to inflate the cuff in all cases, especially in a neonate patient.

When you are inflating a pediatric cuffed ETT, only inflate the cuff with enough air to inflate the pilot cuff, do not over-inflate. This may take less than the normal 10 ml of air in an adult ETT cuff.

References:

Hoffman RJ, Dahlen JR, Lipovic D, Stürmann KM. Linear Correlation of Endotracheal Tube Cuff Pressure and Volume. Western Journal of Emergency Medicine. 2009;10(3):137-139.

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http://www.joacp.org/text.asp?2011/27/3/358/83682. Lichtenthal, PL and Borg, UB. Endotracheal cuff pressure: role of tracheal size and cuff volume. Critical Care. 2011;15(1):147. http://ccforum.com/content/15/S1/P147.

OROGASTRIC (OG) TUBE INSERTION PROTOCOL

Paramedics may use this protocol to facilitate gastric decompression of the intubated patient after endotracheal intubation has been performed and verified, OR with use of the King LTS-D Airway.

Contraindications:
- Known esophageal disease (such as varices).
- Gastric bypass.
- Penetrating neck trauma.

Especially consider use in:
- Submersion injuries (near drowning).
- Intubated Pediatric Patients.
- Any patient who has been aggressively artificially ventilated.

Equipment:
1. 8 or 10 French Sump Catheter.
2. Lubricant (water-soluble).
3. 60 mL catheter tip syringe.
4. Tape.

Procedure:
1. Estimate the length of the tube needed to reach the stomach by measuring the tube from the corner of the mouth to the earlobe and down to the xiphoid process. Mark the length with tape.
2. Lubricate the Salem sump tube with a water-soluble lubricant (e.g. KY Jelly).
3. Insert the tube through the oropharynx OR through the gastric access lumen on the KING LTS-D Airway until the marked depth is reached.
4. If the tube coils in the posterior pharynx, direct laryngoscopy can be utilized to place the tube in the esophagus. If the patient begins to cough, you may be advancing into the trachea. Reattempt insertion…
5. Verify placement (see OG Placement Verification).

Placement Verification:
1. Using a 60 mL catheter tip syringe, instill 10 mL (consider 2-5 mL in children less than one year of age) of air into tube and auscultate over epigastrium for air sounds.
2. Aspirate for gastric contents and assess for cloudy, green, tan, brown, bloody or off-white color contents consistent with gastric contents.
3. Secure tube with tape.

Gastric Decompression:
1. Attach the tube to suction (and suction intermittently as needed). You can also provide continual suction by setting the suction device at 20-60 mmHg negative pressure (TORR).
2. If suction is not readily available, connect the 60 mL syringe to the tube with keeping the (blue) air vent patient. This will allow the sump function of the tube to work until suction can be applied and will also prevent gastric contents from leaking from the tube.

Special Note:
- **If you cannot place the OG tube quickly (no more than 2 attempts), forego the procedure – DO NOT delay transport.**
- **The blue air vent must remain patent to ensure proper sump function and the prevent damage to the gastric lining during suction.**

PEDIATRIC – PAIN MANAGEMENT (TRAUMATIC)

- Assess ABC's are intact, stabilize as necessary.
- Pulse oximetry / Cardiac monitor / ETCO2 NC.

Oxygen as indicated.

Obtain IV access (Critical patients may have IO access).

If there is an obvious fracture, refer to "Fractures General Care" protocol.

For acute traumatic injuries causing **extreme pain:**

- **Fentanyl – 1 mcg/kg slow IV/IO.**

If **no IV access** is available, or patient has only minor injuries:

- Consider Fentanyl – 2 mcg/kg IN (Intranasal) or IM.

Fentanyl may be repeated, titrated to effect, given at least 5 – 10 minutes apart:

- 0.5 mcg/kg via IV/IO.
- 1 mcg/kg via IN or IM.

If patient is allergic to Fentanyl.

Morphine 0.05 mg/kg IV/IO, <u>Max initial dose of 5mg.</u>

If no IV access is available: Consider Morphine 0.1 mg/kg IM

Morphine may be repeated **one time** if needed, at least 5 minutes after initial dose.

Ketamine to assist with pain management is <u>**NOT**</u> to be given to pediatric patients unless a physician orders it.

<u>Use extreme caution when administering narcotics to pediatric patients</u>
USE PEDIATRIC DRUG CALCULATION CHARTS, WHEN POSSIBLE, TO CONFIRM!

Transport as indicated.

THERE ARE <u>NO STANDING ORDERS FOR ANALGESICS IN ABDOMINAL PAIN</u>, OR OTHER MEDICAL COMPLAINTS, CONSULT WITH ON-LINE MEDICAL CONTROL AS NEEDED.

POSITIVE PRESSURE AIRWAY PROTOCOL, part 1 – Pediatric CPAP

CPAP = **Continuous <u>Positive</u> Airway <u>Pressure</u>. CPAP delivers "positive pressure" to patients who have effective ventilation, but have ineffective respiration (exchange of O2 and CO2). Dyspnea patients who present with an altered mental status have likely progressed to respiratory failure. <u>A patient with an altered level of consciousness is not a candidate for CPAP and needs more evasive treatments such as a BVM</u>.**

<u>PEDIATRIC Settings</u>:

(CPAP or PEEP): 2-5 cmH2O, **not to exceed 5 cmH20 unless otherwise directed by a physician / on-line medical control. Ranges exceeding 10 cmH20 should be avoided. For adolescents similar in size of an adult, adult ranges may be considered but should be approved by a physician prior to use.** -- Attach the 2-5 cmH2O valves to the CPAP device as needed. --

<u>CPAP Indications</u>:

For conscious and alert patients with moderate to severe respiratory distress (accessory muscle use or tripod position) from an underlying pathology or acute condition that affects ***respiration*** at the alveolar level.

- Bronchiolitis, Pneumonia or Asthma.
- Near drowning / submersion injuries.
- Pulmonary edema (see notes below):
  - May be seen as an underlying pathology in pediatric patients with special needs conditions.
  - May be seen as an acute presentation where there has been in inhalation injury from toxic gases such as chemical exposure or smoke inhalation on fire scenes.

<u>CPAP Contraindications</u>:

- Respiratory Arrest (needs BVM instead).
- Signs and symptoms of a pneumothorax or chest trauma.
- Active gastrointestinal bleeding or vomiting.
- Patient unable to follow verbal commands or altered mental status.
- Inability to properly fit the CPAP mask and strap or tracheotomy.
- Overdoses.

<u>CPAP Procedure</u>:

1. Assure ABC's are intact, stabilize as needed, monitor ECG, SPO2 and ETCO2.
2. Set CPAP device to desired pressure setting (2-5 cmH2O for pediatrics), titrate as needed.
3. Talk to the patient and/or parents, and explain the procedure / therapy.
4. <u>Once the device is running</u>, then apply to the patient. Use an appropriate size mask for the patient and ensure a good seal as best possible.
5. Constantly re-assess the patient, if their mental status deteriorates, discontinue and ventilate PRN. <u>Nebulized medications such as Albuterol can be interlinked if necessary</u>.

<u>Monitoring CPAP Effectiveness</u>:

Pediatric CPAP monitoring should include cardiac monitoring, ETCO2, pulse oximetry, and frequent assessments of lung sounds, monitor for worsening gastric distension and temperature assessment.

- Improving skin color, mental status.
- Improving respiratory tidal volume, lung sounds.
- Decreasing respiratory rate, accessory muscle use and retractions.
- Decreasing anxiety or agitation.
- A normalizing heart rate.

POSITIVE PRESSURE AIRWAY PROTOCOL, part 2 - PEEP valve use

*Positive End Expiratory Pressure use in **Pediatric** Patients*

PEEP = **Positive End Expiratory Pressure. PEEP delivers "positive pressure" at the end of exhalation to prolong the exhalation time, helping to exhale CO2. This is delivered by attaching a PEEP valve onto a BVM for patients in respiratory failure or arrest who require ventilatory support.** *PEEP may be beneficial; however, it can worsen hypoperfusion, therefore it should NOT be used in this situation or cardiac arrest*.

PEDIATRIC Settings:

(CPAP or PEEP): 2-5 cmH2O, **not to exceed 5 cmH20 unless otherwise directed by a physician / on-line medical control. Ranges exceeding 10 cmH20 should be avoided. For adolescents similar in size of an adult, adult ranges may be considered but should be approved by a physician prior to use.** -- Attach the 2-5 cmH2O valves to the CPAP device as needed. --

PEEP Indications:

Most patients who require artificial ventilation will benefit from PEEP. Physiologic PEEP exists in people upon natural exhalation. PEEP valve use should especially be considered in patients who require BVM ventilation and have underlying pathology that is affecting respiration, such as pulmonary edema, COPD, Asthma, pneumonia, atelectasis, or near drowning patients.

PEEP Contraindications:

- Signs and symptoms of a pneumothorax or chest trauma.
- Tracheotomy (precaution).
- Active gastrointestinal bleeding or vomiting.
- **Avoid use in hypotension or cardiac arrest**.

PEEP Procedure:

- Attach PEEP valve onto exhalation port of BVM and adjust settings by rotating red knob.

PEDIATRIC – REFUSALS (requirements to be met)

In situations where there is considerable mechanism of injury or possibility for medical emergencies with pediatric patients, refusals of transport should be avoided if at all possible.

However, it is recognized that parents or designated caregivers/guardians may elect to transport patients on their own, or even forego further assessment, treatment, or transport. Situations where refusal of transport should be avoided whenever possible are as follows:

- BRUE (Brief Resolved Unexplained Event) / apparent life-threatening event.
- Anaphylaxis / allergic reaction / envenomation.
- Near drowning.
- Altered Mental Status.
- Seizure.
- Possible head injuries.
- Dyspnea / Breathing difficulty.
- Medication overdose or poisoning (including chemical exposure).
- **<u>Any suspected abuse or neglect situation</u>**.

In order for caregivers/guardians to consider refusing EMS transport, EMS providers must be able to determine the following:

- The patient is **alert and oriented** appropriate for their age.
- The patient has **effective** work of **breathing**.
- The patient is hemodynamically **stable.**
- The patient will be left in a **safe** environment.
- The patient will be in the **care** of an appropriate guardian/caregiver.

When possible, a complete set of vital signs shall be obtained prior to making any decision not to transport a pediatric patient. This may also include other diagnostics as appropriate for patient's condition. (Blood glucose, ECG, pulse oximetry, etc.)

If any parent, guardian, or otherwise deemed caregiver is refusing to allow transport against medical advice of EMS personnel, proceed as follows:

- Contact on-duty supervisors immediately.
- Avoid conflict, when possible, involve law enforcement as needed.

PEDIATRIC PATIENT SAFE TRANSPORT GUIDELINES

EMSC
Emergency Medical Services for Children
NATIONAL RESOURCE CENTER

NHTSA
People Saving People

The Do's and Don'ts of Transporting Children in an Ambulance

Approximately 6 million children are transported by emergency medical services (EMS) vehicles each year in the United States. There are risks of injury associated with transport that can be minimized. An ambulance is NOT a standard passenger vehicle. Unlike the well-developed and publicized child passenger safety standards and guidelines, specifications for the safe transport of ill and injured children in ambulances are still under development. Standard automotive safety practices and techniques cannot be applied directly to EMS vehicle environments due to biomechanical and practical differences. Caution is encouraged in the application of passenger vehicle principles to ambulances and in the utilization of new and unproven products.

The Emergency Medical Services for Children (EMSC) Program supports efforts to improve the safety of pediatric patients being transported in EMS vehicles. Through an EMSC grant, the Division of Pediatric Emergency Medicine at Johns Hopkins Children's Center conducted the first ever ambulance vehicle crash safety tests using instrumented crash test dummies to test the performance of these vehicles and a variety of occupant and equipment restraint practices currently available or in use. The data from these and other detailed safety tests are currently being analyzed and will be an integral element in the development of safety guidelines for the transport of ill and injured children.

Currently, the National Highway Traffic Safety Administration's (NHTSA) Office of EMS and Occupant Protection Division are working with the EMSC Program to develop recommendations for how EMS can safely transport ill or injured – and uninjured – children in ambulances.

Until consensus recommendations are available, certain practices can significantly decrease the likelihood of a crash, and in the event of a crash or near collision, can significantly decrease the potential for injury. These practices are listed below in the format of "Do's" and "Don'ts."

Importantly, as is mandated in several states, the NHTSA Emergency Vehicle Operating Course (EVOC) National Standard Curriculum, or its equivalent, is an integral part of this transport safety enhancement.

Do's

- ☑ DO drive cautiously at safe speeds observing traffic laws.
- ☑ DO tightly secure all monitoring devices and other equipment.
- ☑ DO ensure available restraint systems are used by EMTs and other occupants, including the patient.
- ☑ DO transport children who are not patients, properly restrained, in an alternate passenger vehicle, whenever possible.
- ☑ DO encourage utilization of the EVOC, National Standard Curriculum.

Don'ts

- ☒ DO NOT drive at unsafe high speeds with rapid acceleration, decelerations, and turns.
- ☒ DO NOT leave monitoring devices and other equipment unsecured in moving EMS vehicles.
- ☒ DO NOT allow parents, caregivers, EMTs or other passengers to be unrestrained during transport.
- ☒ DO NOT have the child/infant held in the parent, caregiver, or EMT's arms or lap during transport.
- ☒ DO NOT allow emergency vehicles to be operated by persons who have not completed EVOC or an equivalent course.

FAMILY VIOLENCE / CHILD ABUSE PROTOCOL

INDICATORS:

- Fear of household member.
- Reluctance to respond when questioned.
- Unusual isolation, unhealthy, unsafe living environment.
- Poor personal hygiene/inappropriate clothing.
- Conflicting accounts of the incident.
- History inconsistent with injury or illness.
- Indifferent or angry household member.
- Household member refused to permit transport.
- Household member prevents patient from interacting openly or privately.
- Concern about minor issues but not major ones.
- Household with previous violence.
- Unexplained delay in seeking treatment.

Direct questions to ask when alone with patient and time available:

1. Has anyone at home ever hurt you?
2. Has anyone at home touched you without your consent?
3. Has anyone threatened you?
4. Are you afraid of anyone at home?

Signs and Symptoms:

- Injury to soft tissue areas that are normally protected.
- Bruise or burn in the shape of an object.
- Bite marks.
- Rib fracture in the absence of major trauma.
- Multiple bruising in various stages of healing.

Protocol:

1. Patient care is first priority.
2. If possible, remove patient from situation and transport.
3. Summons police assistance as needed.
4. If suspected sexual assault, then follow sexual assault protocol.
5. Obtain information from patient and caregiver.
6. Do not judge.
7. Report suspected abuse to hospital after arrival. Make verbal and written report.

Report Suspected Child Abuse and Neglect

State of Tennessee Phone Number:
1-877-237-0004

Secure On-line site:
https://apps.tn.gov/carat/

National Domestic Violence Hotline:
1-200-799-SAFE (7233)

DEATH OF A CHILD AND SUDDEN INFANT DEATH SYNDROME (SIDS)

There is no normal parental reaction to the death of a child or a SIDS event. Individual responses may range from emotional outbursts to apparent withdrawal. Rescuers should not make any assumptions or judgments. Maintain a professional demeanor at all times. Perform the initial assessment, environmental assessment, and focused history as part of the clinical process. Observe, assess, and document accurately and objectively.

1. Ensure scene safety.
2. Perform a scene survey to assess environmental conditions and mechanism of illness or injury.
3. Form a general impression of the patient's condition.
4. Observe standard precautions.
5. Establish patient responsiveness.
6. Assess airway and breathing. Confirm apnea.
7. Assess circulation and perfusion.
8. Initiate cardiac monitoring. Confirm absent pulse.
9. Determine whether to perform further resuscitation measures: If patient does not exhibit lividity or rigor, proceed with cardiopulmonary resuscitation. During resuscitation, perform steps 11 and 12 below. Initiate transport. If patient exhibits lividity and rigor, do not resuscitate as permitted by medical direction. Proceed with step 10. Note: Lividity can be mistaken for bruising and evidence of abuse. Be careful not to make any assumptions or judgments.
10. Provide supportive measures for parents and siblings:
    - Explain the resuscitation process, transport decision, and further actions to be taken by hospital personnel or the medical examiner.
    - Reassure parents that there was nothing they could have done to prevent death.
    - Allow the parents to see the child and say goodbye.
    - Maintain a supportive, professional attitude no matter how the parents react.
    - Whenever possible, be responsive to parental requests. Be sensitive to ethnic and religious needs and make allowances for them.
11. Obtain patient history using a nonjudgmental approach. Ask open-ended questions as follows:
    - Has the child been sick?
    - Can you describe what happened?
    - Who found the child? Where?
    - What actions were taken after the child was discovered?
    - Has the child been moved?
    - When was the child last seen before this occurred, and by whom?
    - How did the child seem when last seen?
    - When was the last feeding provided?
12. Reassess the environment. Document findings, noting the following:
    - Where the child was located upon arrival?
    - Description of objects located near the child upon arrival?
    - Unusual environmental conditions, such as a high temperature in the room, abnormal odors, or other significant findings?

CONTINUED ON NEXT PAGE…

DEATH OF A CHILD AND SUDDEN INFANT DEATH SYNDROME (SIDS)

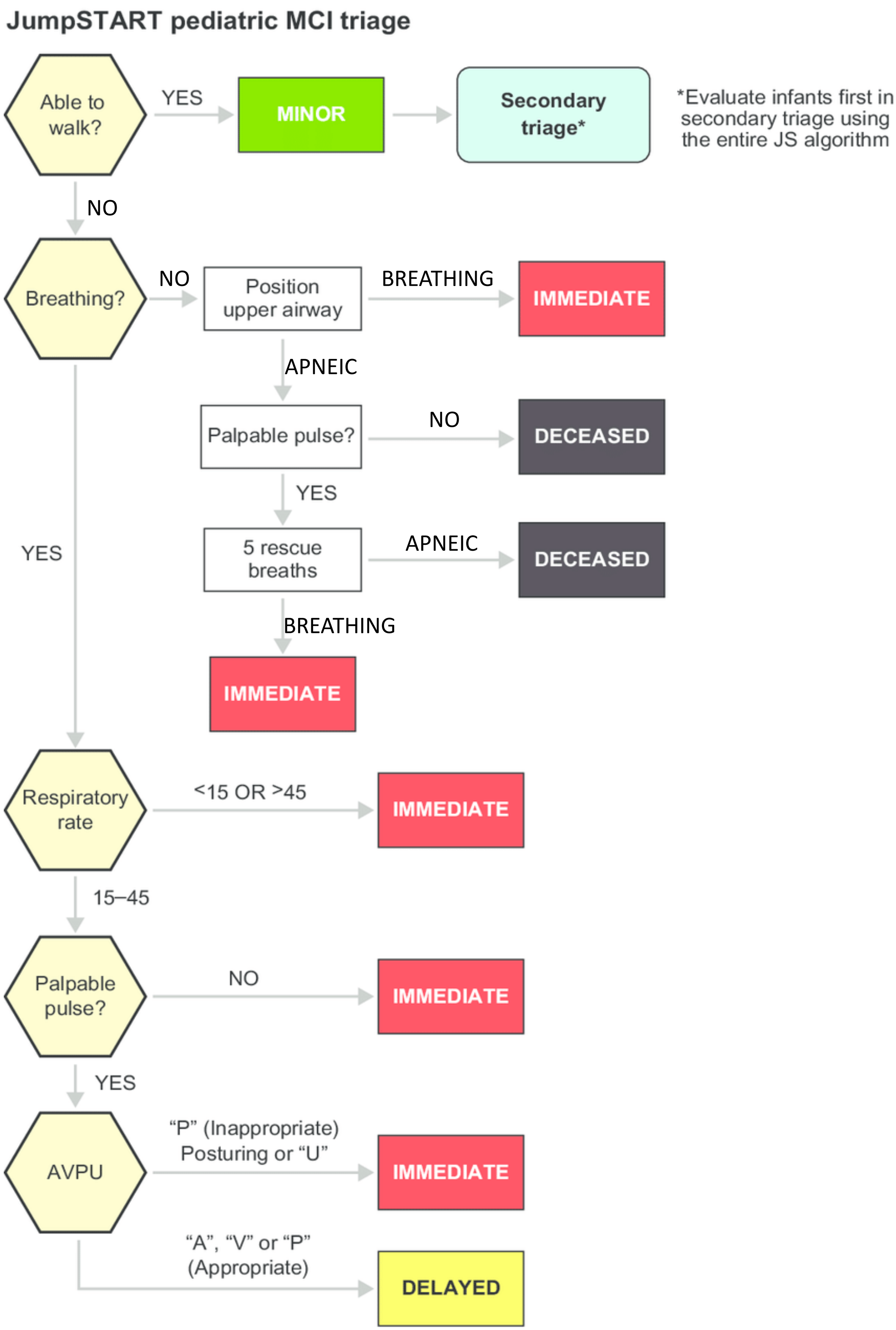
(CONTINUED)

13. If the parents interfere with treatment or attempt to alter the scene, initiate the following actions:
    - Remain supportive, sympathetic, and professional.
    - Avoid arguing with the parents or exhibiting anger.
    - Do not restrain the parents or request that they be restrained unless scene safety is clearly threatened.
14. Document the emergency call, including the following information:
    - Time of arrival.
    - Initial assessment findings and basis for resuscitation decision.
    - Time of resuscitation decision.
    - Time of arrival at hospital if resuscitation and transport were initiated.
    - Parental support measures provided if resuscitation was not initiated.
    - History obtained (note who provided the information).
    - Environmental conditions.
    - Time law enforcement personnel arrived on scene.
    - Time that scene responsibility was turned over to law enforcement personnel.

The priority of emergency medical services personnel on scenes involving infant/pediatric death is to provide expeditious transport and deliver emergency medical treatments. It is the primary role of law enforcement and medical examiners to perform detailed assessments of the scene, environment, and events surrounding the incident.

EMS personnel are not expected to perform any other duties from delivering assessment, treatment, and transport. Any additional information acquired during these phases of the call shall be documented appropriately, otherwise EMS shall not delay treatment or transport to gather information about the scene.

PEDIATRIC MCI TRIAGE - JumpSTART



PEDIATRIC – VENTRICULAR FIBRILLATION / PULSLESS V-TACH

Per the 2015 & 2020 AHA Guidelines, you no longer have to perform CPR prior to defibrillation.

Focus on HIGH QUALITY CPR:
- Compressions at 100-120 / min.
- Compress 1/3 to 1/2 depth of chest wall.
- Allow adequate chest recoil.
- Minimize interruptions less than 10 secs.

Defibrillate at 2 Joules/Kg ASAP
(Joule setting can be the child's weight in lbs.)
Repeat defib every 2 minutes AS NEEDED at 4 Joules/Kg
(Joule setting can be twice the child's weight in lbs.)
Reassess every 2 minutes and repeat Defibrillation PRN

RESUME CPR

Insert OPA (prn) and Ventilate with BVM attach to high flow O2.

Can patient be effectively ventilated with BVM and oral airway?

NO

YES

Establish Vascular Access (IO, if faster)

Intubate patient ASAP and attach mainstream ETCO2 to assess presence of waveform Capnography.

Administer **Epinephrine (1:10,000) 0.01 mg/kg IV/IO:**
- Repeat every 3 – 5 minutes, no max dose, until ROSC or resuscitation is terminated.

Intubate if patient remains apneic, following CPR, Defib, and Meds.
Intubation is a priority if you cannot initially ventilate with basic interventions.

Administer **Amiodarone 5 mg/kg IV/IO**:
- Repeat ONCE in 3 – 5 minutes at **2.5 mg/kg IV/IO** if patient still in shockable rhythm. If Amiodarone is unavailable, Lidocaine 1 mg/kg IV/IO bolus to a maximum of 3 mg/kg can be given.

TRANSPORT EMERGENCY TO THE NEAREST FACILITY

- Continue CPR with pulse and rhythm checks every 2 minutes (5 cycles).
- Consider Magnesium Sulfate 25-50 mg/kg IV/IO (max of 2 grams) over 1-2 mins if Torsades de Pointes is present or if the patient is malnourished.
- Pediatric Narcan - 0.01 mg/kg IV, IO, IM or IN maximum of 2 mg per dose. May repeat once.

Consider Irreversible Causes and Treat as Indicated:

Hypoxia	Toxins
Hypovolemia	Trauma
Hydrogen Ion	Tension Pneumothorax
Hypo/hyperkalemia	Tamponade
Hypothermia	Thrombosis

PEDIATRIC – ASYSTOLE / PULSELESS ELECTRICAL ACTIVITY

- Compressions at 100-**120** / min.
- Compress at least 1/3 to 1/2 depth of chest.
- Allow adequate chest recoil.
- Minimize interruptions, no more than 10 secs without compressions.

Insert OPA (prn) and Ventilate with BVM attached to high flow O2.

Can patient be effectively ventilated with BVM and oral airway?

NO

Intubate patient ASAP and attach mainstream ETCO2 to assess presence of waveform capnography.

YES

Establish Vascular Access (IO, if faster)

Administer **Epinephrine (1:10,000) 0.01 mg/kg:**

- Repeat every 3-5 minutes, no max dose, until ROSC or resuscitation is terminated.

TRANSPORT EMERGENCY TO THE NEAREST FACILITY

- Continue CPR with pulse and rhythm checks every 2 minutes (5 cycles)
- Pediatric Narcan – 0.01 mg/kg IV, IO, IM or IN maximum of 2 mg. May repeat once.
- Waveform Capnography with a reading of less than 10 mmHg may indicated poor CPR or suggest consultation of on-line medical control to terminate resuscitation efforts.
- Persistent Asystole despite >20 minutes of resuscitation may also suggest a consult with on-line medical control to consider termination of efforts.

Consider Irreversible Causes and Treat as Indicated:

Hypoxia	Toxins
Hypovolemia	Trauma
Hydrogen Ion	Tension Pneumo
Hypo/hyperkalemia	Tamponade
Hypothermia	Thrombosis

PEDIATRIC – SYMPTOMATIC BRADYCARDIA

- Assess ABC's, stabilize as needed.
- Pulse oximetry, cardiac monitor, waveform capnography can help assess perfusion status.
- Work to obtain IV access, refer to vascular access protocol if needed (for IO infusion).
- If patient is stable, acquire 12-Lead ECG every 10 minutes throughout transport.

Unstable Patient?

Bradycardia with patient showing the following signs:

- Critical Hypotension.
- Altered Mental Status.
- Unresponsive.
- Ischemic chest discomfort.

Stable Patient?

- Symptomatic, however not yet critical.

YES

Initiate **CPR** in a unstable patient:

- **Infant** with **HR < 80 bpm**.
- **Child** with **HR < 60 bpm**.

Establish Vascular Access:

- IV is preferred if stable.
- IO may be initial attempt if unstable.

Administer **Epinephrine** and repeat every 3-5 minutes as needed:

- Administer **0.01 mg/kg 1:10,000** IV or IO with a maximum dose of 1 mg.
- Administer **0.1 mg/kg 1:1,000** ET tube if IV/IO is not available.

If no response to O2 and Epinephrine give:

Atropine 0.02 mg/kg IV/IO

- Minimum dose of 0.1 mg.
- Maximum dose 0.5 mg per dose.
- May repeat once in 3 – 5 minutes.

Transport emergency traffic.

NOTE:
If organophosphate poisoning is suspected as being the cause of the bradycardia, administer 0.05 mg/kg of Atropine IV (usual dose 1-5 mg), may be repeated in 5 to 15 minutes.

PEDIATRIC TACHYCARDIA – WIDE COMPLEX WITH PULSE (V-TACH)

- Assess ABC's, stabilize as needed.
- Pulse oximetry, cardiac monitor, waveform capnography can help assess perfusion status.
- Work to obtain IV access, refer to vascular access protocol if needed (for IO infusion).
- If patient is stable, acquire 12-Lead ECG to confirm V-tach prior to treating.

Unstable Patient?

- Altered Mental Status.
- Mottled skin / cyanosis.
- Hypotension as defined as systolic BP less than 70 + (age in years' x 2)

YES

If the patient is conscious / responsive to painful stimulus, give:

- **Versed 0.1 – 0.2 mg/kg IV/IO**, if no IV/IO is available, consider Versed 0.2 mg/kg IM
- Max dose 2 mg IV or 5 mg IM unless otherwise directed by on-line MD.

Proceed to electrical therapy / synchronized Cardioversion:
Place defibrillator in synchronized made and shock in the following sequence until patient converts:

0.5 J/kg → 1 J/kg → 2 J/kg (max 2 J/kg)

Stable Patient?

- Symptomatic, however not yet critical.

Obtain IV / IO access if indicated.

If patient is clinically stable, yet symptomatic:

- Administer **Amiodarone 5 mg/kg mixed in 100 mL bag of Normal Saline infused** (Max 150 mg) **slowly over 20 minutes.**
- If Amiodarone is unavailable, Lidocaine 1 mg/kg IV can be given, may repeat in 5-10 minutes at 0.5 mg/kg IV.

Refer to the Amiodarone drip instructions in the formulary section if needed.

Re-assess and treat appropriately, consult on-line medical control if further orders are needed.

TRANSPORT WITHOUT DELAY TO THE NEAREST FACILITY.

PEDIATRIC TACHYCARDIA – NARROW WITH A PULSE

- Assess ABC's, stabilize as needed.
- Attach pulse oximetry and cardiac monitor.
- Work to establish **vascular access** as soon as possible.
- 12-Lead ECG if applicable to size of patient, transmit.

Is the Patient stable?

NO

Unstable Patient?
Acutely tachycardic heart rate (> 180 –child or > 220 – infant), or the following:

- Hypotension.
- Altered Mental Status.
- Acute heart failure.
- Ischemic chest discomfort.

YES

If the patient is conscious / responsive to painful stimulus, give:

- **Versed 0.1 mg/kg IV/IO**, if no IV/IO is available, consider Versed 0.2 mg/kg IM.

Proceed to electrical Cardioversion:
Place defibrillator in synchronized made and shock in the following sequence until patient converts:

0.5 J/kg → 1 J/kg → 2 J/kg (max 2 J/kg)

YES

If patient is stable, refer to these 4 considerations before treating the tachycardia as SVT:

- **HR >180/min in a child, or > 220/min for infant.**
- R-R intervals are regular.
- The width of the QRS at its base must be less than or equal to 1mm (1 small block).
- **Is there any underlying history suggesting a compensatory tachycardia needing fluids?**

For stable SVT, attempt **vagal maneuvers:**

If these are ineffective:

1. Begin recording ECG strip...
2. Administer **Adenosine 0.1 mg/kg IV** (maximum of 6 mg) with rapid NS flush.
3. May repeat Adenosine 0.2 mg/kg IV (maximum of 12 mg) once with rapid NS flush.
4. Total maximum dose of 18 mg.

Re-assess and treat appropriately, consult on-line medical control if further orders are needed.

TRANSPORT WITHOUT DELAY TO THE NEAREST FACILITY.

NOTE:
*Most pediatric tachycardias are **compensatory in nature**, secondary to dehydration / hypovolemia, and respond best to **IV fluid therapy**. If you are unsure of the best treatment approach, consult with on-line medical control whenever possible.*

PEDIATRIC DYSPNEA – UPPER AIRWAY OBSTRUCTION

*Foreign Body Airway Obstruction (FBAO) / **Choking***

Airway obstruction may quickly lead to hypoxia, that will lead to anoxic brain injury or cardiac arrest. <u>BLS skills shall be applied immediately</u> while the paramedic prepares for more invasive interventions (Cricothyrotomy)

Is the patient **able** to cough, speak, or breathe?

<u>NO</u>, they can NOT move air

YES

Continue to monitor and transport as indicated.

Apply Basic Life Support Skills

Child
> 1 year old

Give **5 back slaps and 5 abdominal thrusts** with one hand while supporting the patient from behind.

Infant
< 1 year old

Hold the baby carefully, with head slightly downward... **give series of 5 back slaps followed by 5 chest thrusts** until the object is removed.

<u>If the patient becomes unresponsive, begin high quality CPR…</u>

- Look in the mouth before giving breaths to assess for the object.
- Do not perform blind finger sweeps, only sweep to remove a visible object.

Move to the Needle Cricothyrotomy Protocol as needed.

<u>Paramedics may attempt to visualize the obstruction via laryngoscopy, and attempt to remove with forceps.</u>

- Caution should be applied NOT to further advance the obstruction into the trachea/airway.

Continue to monitor and transport emergency.

PEDIATRIC DYSPNEA - UPPER AIRWAY OBSTRUCTION

CROUP or EPIGLOTTITIS *(STRIDOR NOTED)*

- Assure ABC's are intact, stabilize as necessary.
- Pulse oximetry / Cardiac monitor/Side-stream ETCO2 should be used.

↓

Oxygen to keep O2 sats > 90%.

↓

If **stridor**/croup give **humidified O2**, (3-4 mL's of Normal Saline @ 8 lpm in Nebulizer)

↓

If **stridor**/croup, give **nebulized Epi** 1:1,000 (Mix 1 vial (1 ml) of Epinephrine 1:1,000 (1 mg) mixed with 3-4 mL's of NS) for ages < 5 years.

- Deliver via nebulizer at 6-8 LPM flow rate from O2 regulator.
- Deliver via blow-by method if the child will not tolerate a mask to receive the nebulized medication.
- ***In cases of respiratory failure (evidenced by altered mental status or other signs of fatigue) from upper airway obstruction, you can interlink the nebulizer into a BVM to provide assist ventilations as needed***.

↓

If no change in patient condition, supplement ventilations with BVM and intubate as needed.

↓

If allergen exposure, go to the anaphylaxis protocol.

Transport as indicated.

Respiratory patients should be positioned upright when possible.

When treating a small child with suspected Croup or Epiglottitis, try to **keep the patient calm**. If the child becomes increasingly upset or cries more, they could worsen their airway edema. You would be within standards of care and protocol to forego pain or anxiety inducing procedures in children such as BP assessment, blood glucose assessment, and IV therapy unless these procedures are absolutely necessary. A priority in cases of Croup or Epiglottitis is to calm the child, administer the nebulized Epi as indicated and transport without delay.

PEDIATRIC DYSPNEA - LOWER AIRWAY OBSTRUCTION

*SUSPECTED **ASTHMA** / WHEEZING NOTED*

- Assess ABC's, stabilize as necessary.
- Pulse oximetry / Cardiac monitor / Side-stream **ETCO2 should be used**.

↓

Oxygen to keep O2 sats > 90%, use BVM to support breathing if respiratory failure is noted (AMS).

↓

If patient has a history of **Asthma, presents with wheezing** or poor air movement, then give:
Albuterol 2.5 mg in 3 mL's via nebulizer (may be interlinked with BVM as needed).
***** If Albuterol is not available due to shortages, then DuoNeb can be used. *****

↓

ONLY IF patient is ALERT still, Consider CPAP, initiated at 2-5 cmH20.
Use appropriately sized mask, Reference CPAP protocol if needed.

↓

If in extremis, or if it is a possible allergen exposure, consider 0.01 mg/kg Epi 1:1,000 IM in thigh.

↓

Obtain IV access (may give one nebulizer treatment without IV access).

↓

Magnesium Sulfate –20 mg/kg (maximum of 2 grams) to be mixed in a 100-150 mL bag of Normal Saline infused over 10 minutes **if severe difficulty breathing** (minimum weight of 10 kg).

- Amount of magnesium sulfate (packaged as 5 grams/10 mL) is 1 mL per 10 kg

↓

Solu-medrol 1 mg/kg IV or IM (maximum of 125 mg).

↓

Repeat Albuterol only in 10 minutes if an IV is successfully established.

- In the absence of IV access, contact on-line medical control for orders to proceed with additional Albuterol treatments.

↓

If no change in patient condition, supplement ventilations with BVM and intubate as needed.

↓

If allergen exposure, go to the anaphylaxis protocol.

Transport as indicated.

Respiratory patients should be positioned upright when possible

PEDIATRIC - SHOCK PROTOCOL (all types)

- Assure CAB's.
- Pulse oximetry.
- Oxygen via NRB.
- Cardiac monitor.

Place in supine position as tolerated.

Obtain immediate vascular access.

Give 20 mL/kg LR bolus (if LR is unavailable, use NS as fluid):

- May be repeated PRN.
- Check lung sounds after each bolus.
- Liver engorgement may indicate too much fluid, too fast (swelling to RUQ of ABD).
- Use 10 gtt/ml tubing, or other MACRO drip systems for any bolus infusions.

Attempt to determine the etiology of shock by history and exam!

Hypovolemic – Hemorrhagic Shock:

- Continue with IV fluid bolus as necessary and titrate to effect to maintain stable perfusion based on the patient's condition (medical or trauma needs), minimally acceptable systolic BP: (>70 + 2 x age in years).

Anaphylactic Shock:

- Continue with IV fluids bolus and go to Anaphylaxis / Allergic Reaction Protocol.

Cardiogenic Shock:

- Go to appropriate protocol.
- After the rate and rhythm normalize and the patient is still in shock, then start an Epinephrine Drip at 0.1 – 1 mcg/kg/min, carefully titrated to effect. Contact on-line medical control if further orders are needed Continue with fluid bolus and go to Anaphylaxis / Allergic Reaction - Anaphylaxis protocol.

Spinal Shock (Neurogenic):

- ***If the patient has possible multi-system trauma, withhold vasopressors and consult on-line physician.***
- Begin Epinephrine Drip at 0.1 mcg/kg/min and titrate to effect not to exceed 1 mcg/kg/min.

Septic Shock:

- Initiate **fluids** at 30 mL/kg.
- Move to vasopressors after 30 mL/kg if there is no change, **Epinephrine Drip at 0.1 mcg/kg/min,** carefully titrate to effect, not to exceed 1 mcg/kg/min without on-line physician orders.
- Notify the receiving facility of a possible **sepsis alert** patient.

Transport Emergency.

PEDIATRIC - ANAPHYLAXIS / ALLERGIC REACTION

- Assess ABC's, stabilize as needed.
- Pulse oximetry.
- Oxygen via Non-rebreather.
- Cardiac monitor.

Patients <18 years old

Administer Epinephrine (1:1,000):
0.01 mg/kg IM, maximum each dose 0.3 mg.
May repeat once after 15 minutes.

IM injection is preferred in the THIGH for peds.

Give **Albuterol** 2.5 mg in 3 mL of NS nebulized, if wheezing or dyspnea is present.

Use waveform capnography to assess for bronchospasms PRN (shark-fin waveform).

*** If Albuterol is unavailable due to shortages, then a DuoNeb can be used. ***

Obtain IV access (vascular access).

If hypotensive or inadequate tissue perfusion:
- Administer Normal Saline 20 mL/kg.

Administer Antihistamine:

H-1 Blocker
Administer **Benadryl (Diphenhydramine)**
1 mg/kg IV/IO, max dose of 25 mg.
- If unable to obtain IV access in pediatrics: Benadryl (1 mg/kg) up to 25 mg deep IM.

Administer Solumedrol:

- Pediatric 1 mg/kg IV or IM (maximum of 125 mg)

If patient is still in extreme Anaphylaxis after treatment above, then consider Epinephrine drip (see Epi Drip in medication section).

Epi drip @ 0.1 - 1 mcg/kg/min, titrate to effect
OR
Consult on-line MD for repeat Epi IM doses.

Signs of extreme/persistent anaphylaxis:
- Profound hypotension (shock).
- Dyspnea.
- Stridor or wheezing.

Transport as indicated.

BRIEF RESOLVED UNEXPLAINED EVENT (BRUE)

Presentation:
An episode in an infant or child less than 2 years' old that is frightening to the observer and is characterized by some combination of the following:

- Apnea (central or obstructive).
- Skin color change: cyanosis, erythema (redness), pallor, plethora (fluid overload).
- Marked change in muscle tone.
- Choking or gagging not associated with feeding.
- A witnessed foreign body aspiration.

- MOST PATIENTS WILL APPEAR STABLE AND EXHIBIT A NORMAL PHYSICAL EXAM UPON ASSESSMENT BY RESPONDING FIELD PERSONNEL. HOWEVER, THIS EPISODE MAY BE THE SIGN OF UNDERLYING SERIOUS ILLNESS OR INJURY.
- **FURTHER EVALUATION BY MEDICAL STAFF IS REQUIRED AND IT IS ESSENTIAL TO TRANSPORT ALL PATIENTS WHO EXPERIENCED AN ACUTE LIFE-THREATENING EVENT (ALTE).**

- Assess ABC's, stabilize as needed,
- Pulse oximetry, ETCO2 monitoring as indicated.
- Cardiac Monitor is required.
- Assess Carbon Dioxide (CO) levels is exposure is suspected.

↓

Perform an initial assessment utilizing the Pediatric Assessment Triangle.

↓

Obtain a description of the event including nature, duration, and severity.

↓

Obtain a medical history with emphasis on the following conditions:

- Known chronic diseases.
- Evidence of seizure activity.
- Current or recent infections.
- Gastroesophageal reflux.
- Recent trauma.
- Medications (current or recent).

→

Oxygen and airway maintenance appropriate for the patient's condition.

↓

Be prepared to assist with ventilation if this type of episode occurs again during transport.

↓

Assess environment for possible causes.

←

IV/IO access as indicated, ONLY if fluids or meds are required.

↓

Transport without delay.

IF THE PARENT OR GUARDIAN REFUSES MEDICAL CARE OR TRANSPORT, CONTACT ON-LINE MEDICAL CONTROL:

- NOTIFY SUPERVISORS ASAP.
- INVOLVE LAW ENFORCEMENT AS NECCESSARY.

PEDIATRIC - BURNS

Stop the Burning!

- Remove burned or smoldering clothes.
- Cool with cool (not cold), moist, sterile towels if available.
- Burns involving more than 10 percent body surface area should be covered with a dry sterile dressing, preserve heat loss when possible.

Remove dry chemicals by brushing off the substance, and remove liquid chemicals by flushing with large amounts of water unless contraindicated according to the ERG handbook.

↓

Assess ABC's, stabilize as needed.

↓

Oxygen via NRB and control airway as indicated.

↓

Cardiac monitor as indicated.

↓

Obtain immediate vascular access.

↓

Is the patient hypotensive?

- **< 70 + (2 x age) in peds.**

NO → See pain management protocol

YES → Only if patient is hypotensive, **Initiate a LR bolus** of 20 mL/kg in pediatrics.

↓

See **pain management** protocol as needed, Fentanyl may be given - 1 mcg/kg
Use caution if hypotensive

BE CAREFUL USING THE "BROSELOW" TAPE... it will give RSI dosing for Fentanyl.

↓

Transport as indicated...Critical burns, which likely require a burn center (Vanderbilt) would be:

- Burns with > 10% BSA partial thickness involvement or worse in pediatrics.
- Any burns that involve the airway or thoracic region (would affect breathing).
- Burns affecting the genitalia "1%".
- **Keep patient warm**, hypothermia is a complication of critical burned patients.
- Focus to prevent infection, use dry sterile dressings (burn sheets) if critical.

PEDIATRIC - DIABETIC EMERGENCY / HYPOGLYCEMIA

- Assess ABC's, stabilize as necessary.
- Obtain vital signs
- Pulse oximetry.

Oxygen as indicated.

Cardiac monitor.

Determine glucose level.

Patient AAOX4, yet symptomatic…see note below.

BG < 70 (less than).

Establish IV access.

Administer Dextrose per the following guidelines:

- D50% 1-2 ml/kg for patients > 8 years old.
- D25% 2-4 ml/kg for patients 6 months to 8 years of age.
- D10% 2-4 ml/kg for patients neonate to 6 months of age.
    - Max. rate of infusion 2 mL/kg/min.

Dextrose may be repeated ONCE only, consult on-line medical control for further orders.

- Repeat in 5-10 minutes if no change in mental status

If D25 or D10 are not available, utilize a syringe of D50:

- To make D25, expel 25 ml of D50 and draw up 25 ml of NS.
- To make D10, expel 40 ml of D50 and draw up 40 ml of NS.

Reminder: IO is appropriate after 2 failed IV attempts or 90 seconds.

If patient is awake, alert, cooperative, and blood glucose is > 50 or an IV cannot be obtained, then oral glucose 15 grams (1 tube) may be given instead of IV Dextrose. (ex: hypoglycemic infants who are stable, apply a small amount of oral glucose to a pacifier and allow the child to consume)

****** DO NOT give anything PO (by mouth) to any patient who has altered mental status. ******

Transport as indicated:

- **Paramedics should be the attending provider** with pediatric patients receiving medications.
- If unable to give Dextrose and the patient's mental status is abnormal, then transport emergency.

BG > 70 (more than).

Transport as indicated.

HYPERTHERMIA / HEAT RELATED ILLNESS

Immediate cooling has been proven to be more beneficial if done prior to transport from sporting events. If athletic trainers have cooling capabilities, allow them to cool the patient prior to transport.

↓

- Assess ABC's, stabilize as needed.
- Assess temperature.
- Cardiac Monitor.

↓

If history is suggestive of heat exhaustion or heat stroke:

- Remove to cooler environment.
- Cool with moist sheets slowly so that the patient will not start to shiver.

↓

Oxygen as indicated.

↓

Obtain IV access and administer Lactated Ringers (LR)

- <60 days: 10 mL/kg.
- >60 days: 20 mL/kg.

↓

If seizures are present, then go to the Seizure protocol.

↓

Transport as indicated.

HYPOTHERMIA

- Assess ABC's, stabilize as needed.
- Assess temperature.

↓

- Remove all wet clothing.
- Protect against heat loss and wind chill.
- Avoid rough and excessive movement.

↓

Monitor cardiac rhythm:
- Treat only life-threatening arrhythmias.

↓

Administer oxygen and begin external warming.

↓

Obtain IV/IO access with NS from fluid warmers:
- <60 days: 10 mL/kg.
- >60 days: 20 mL/kg.

↓

If narcotic ingestion is possible, give Narcan 0.4 – 4 mg IV/IO.

↓

If no pulse or breathing:
- Start CPR.
- Resuscitate per PALS protocol with the following exceptions:
  - Defibrillate **1** time and then **NO MORE**.
  - Lidocaine may or may not work.
- In **pulseless V-Tach, V-Fib with Hypothermia**.
  - Administer **Magnesium Sulfate 20mg/kg IV/IO over 10 minutes**.

↓

Transport as indicated.

- Keep the doors of the ambulance CLOSED with heat on high, especially in the winter, 3 layers of blankets are recommended for re-warming, with at least one blanket under the patient between them and spine boards, etc.

- Note that with a hypothermia as subtle as 96 degrees. The body can lose the ability to form clots / slow bleeding in trauma patients.

NAUSEA / VOMITING

- Assess ABC's, stabilize as needed.
- Control airway and be prepared to suction.
- Pulse oximetry.

↓

Give oxygen as indicated.

↓

Cardiac Monitor, A patient who has been experiencing emesis may be hypovolemic and have electrolyte depletion as well.

- Acquire and transmit a 12 Lead ECG if you suspect any cardiac related illness.

↓

Obtain IV access, administer fluids as needed for dehydration 20 mL/kg LR (NS may be used as an alternative if LR is unavailable) bolus, reassess and repeat as needed to restore normotensive BP.

↓

Administer Zofran:

- > 40 kg: 4 mg IV (Maximum of 4 mg).
- < 40 kg: 0.1 mg/kg IV (Maximum of 4 mg).

↓

Transport as indicated.

PEDIATRIC - OVERDOSE (GENERAL / MEDICATIONS)

- Assess ABC's, stabilize as needed.
- Pulse oximetry / Cardiac monitor/ use Capnography as indicated.

Suction as needed. → Oxygen via NRB. → Aggressive airway control with ventilation if needed.

↓

Intubate as needed.

←

Obtain IV access.
Any hypotension, then given a fluid bolus of isotonic crystalloid (NS or LR) 20 ml/kg in pediatrics.

↓

Check blood glucose, if < 70 or > 400, go to the appropriate hypo/hyperglycemia protocol.

→

If the patient is seizing, then go to the Seizure protocol.

↓

Obtain history:

- Type and amount of poison.
- If possible, bring the container with the patient.
- Route of intake.
- Time of intake.
- History of drug or alcohol usage.

←

If the patient is agitated and a possible stimulant overdose is suspected:
Consider **Versed** 0.05 mg/kg IV/IO (Max. 2 mg IV/IO) or 0.2 mg/kg IM (Max. 5 mg IM)

- Repeat dose if needed in 10 minutes and titrate to effect.

↓

If a **Narcotic Opiate** overdose is suspected:

- Pinpoint pupils.
- Hypotension.
- Decreased Respirations.

Administer **Narcan** 0.01 mg/kg IV, IO, IM or IN not to exceed 2 mg per dose. May repeat once in 5 minutes.

If only Intranasal Narcan 4 mg is available, this may still be used in cases of extremis.

Otherwise, Narcan should be given IM in the absence of vascular access.

If a **Beta Blocker** overdose is suspected and the patient is bradycardic and/or hypotensive:

- Give **Glucagon** 1 mg IVP.

If a **Tricyclic overdose** is suspected AND the patient is unstable (hypotensive, unresponsive: *(Contact medical control if not sure of the drug)*

- Give **Sodium Bicarb 4.2%** 1 mEq/kg IV.

If a **Calcium Channel Blocker** overdose is suspected and the patient is bradycardic and/or hypotensive:

- Give **Calcium Chloride** 20 mg/kg mixed in 100 mL bag of NS and given over 10 minutes.

↓

Transport as indicated.

When in doubt, call online medical control or TN Poison Control Hotline:

1-(800)-222-1222 or 1-(888)-222-1222

https://www.vumc.org/poisoncenter/

PERMISSIVE HYPOTENSION / TRAUMA FLUID RESUSCITATION

Do not delay transport to access IV's unless transport will be delayed due to extrication or patient is entrapped.

Signs of Shock:

- Tachycardia
- Hypotension
- ETCO2 < 30 mmHg.

Administer fluid bolus **to maintain systolic pressure 70 + (2 x age in years)** or maintain peripheral pulses…

- <60 days: 10 mL/kg
- >60 days: 20 mL/kg

Watch for ETCO2 to increase first as initial sign of improvement, then assess for BP to increase…**DO NOT CONTINUE FLUIDS once you reach goal Systolic BP**, you may repeat PRN to maintain target Systolic BP.

Normal Vital Signs Present:

- Not Tachycardic.
  - Infant: 100-180
  - Toddler: 100-140
  - School age: 75-120
- Normotensive Systolic BP.
  - 70 + (2 x age in years)

Reassess every 5 minutes with high MOI patient's, establish IV access PRN to be ready to treat if S/S of shock develop.

DON'T GIVE FLUID BOLUS UNLESS S/S OF SHOCK ARE PRESENT… you may see tachycardia before hypotension.

Consideration:

- Ensure IV fluids are warm when infusion into trauma patients.
- You may consider use of IO device unless area of injury prohibits IO placement when you are unable to obtain peripheral IV's…. fluid administration and rate are the same with IV/IO.
- **Lactated Ringers (LR) is the preferred IV fluid of choice for all patients receiving bolus of isotonic crystalloids.**

Humeral head IO access is NOT permitted for pediatric patients less than 8 years old, unless authorized by on-line medical control.

POISONING / CHEMICAL EXPOSURE / HAZ-MAT / NERVE AGENTS

Personnel safety is the highest priority. **Do not handle the patient unless they have been decontaminated.**
All EMS treatment should occur in the Support Zone after decontamination.

Suction as needed. → Oxygen via NRB. → Aggressive airway control with ventilation if needed. → Intubate as needed. →

Obtain IV access:
Any hypotension, then give 20 mL/kg in peds.

↓

Check blood glucose, if < 70 or > 400, go to the appropriate hypo/hyperglycemia protocol. →

If the patient is seizing, then go to the seizure protocol.

↓

Obtain history:

- Type and amount of poison.
- If possible, bring the container with the patient.
- Route of intake.
- Time of intake.
- History of drug or alcohol usage.

←

If the patient is agitated and a possible stimulant overdose is suspected:

Consider **Versed** 0.05 mg/kg IV/IO (Max. 2 mg IV/IO) or 0.2 mg/kg IM (Max. 5 mg IM)

- Repeat dose if needed in 10 minutes and titrate to effect.

↓

If inhaled poison, remove patient from the source using appropriate PPE / SCBA preferred.
Assess CO levels if exposure to CO is suspected. Consult with / **use Haz-Mat personnel when appropriate.**

↓

If the chemical is a dry substance, then brush off the chemical before irrigating. →

Irrigate with copious amounts of water and reassess for hypothermia.

↓

Transport as indicated.

↓

When in doubt, call online medical control or TN Poison Control Hotline:
1-800-222-1222

For organophosphate / nerve agent poisoning:
Administer **Atropine 0.05 mg/kg** IVP every 5-15 min to dry secretions.

Depending on S/S, administer Nerve Agent Antidote kit:
Mild (Increased secretions, pinpoint, general weakness):

- Decontamination, supportive care.

Moderate (Mild symptoms and respiratory distress):

- 1 Nerve Agent Antidote kit.
- May be repeated in 5 minutes, PRN.

Severe (unconsciousness, convulsions, apnea):

- 3 Nerve Agent Antidote kits.

PEDIATRIC - SEIZURES

Primary Assessment:

- Assess ABC's, stabilize as needed.
- Protect patient from injury.
- Suction as needed.
- Nasal Airway (NPA) as needed.
- **Give O2 AND Assist Ventilations as needed.**

Secondary Assessment:

- Assess vital signs ASAP.
- Cardiac monitor as indicated.
- Pulse oximetry.
- Capnography (required if giving Versed).
- Assess temperature as indicated.

Is patient actively seizing?

YES

Is an IV established?

YES

Give **Versed 0.1 mg/kg IVP**, max dose 2 mg IV/IO.

NO, or not yet…

Give **Versed 0.2 mg/kg IM**, max dose 5 mg IM.

NO

Determine blood glucose level:
If blood glucose is < 70 mg/dl, follow the hypoglycemia protocol.

If the patient continues to have seizures:

IV route – Versed 0.1 mg/kg IV may be repeated 2-3 minutes after the initial dose.

- **Versed IVP may be repeated 2 times, after initial IV or IM doses.**

IM route – Versed 0.2 mg/kg IM may be repeated 5 minutes after the initial dose.

- **Versed IM may only be repeated ONE TIME.**

Transport as indicated.

Special Attention should be applied to ensure that pediatric patients with seizures have an effective work of breathing. "Disordered Control of Breathing" can occur easily due to prolonged or acute seizure activity. If you suspect ineffective breathing due to seizures in a child or infant, provide assisted ventilations with a BVM. Use ETCO2 monitoring to assess for signs of hypercapnea, indicating hypoventilation. (ETCO2 >45)

SELECTIVE SPINAL IMMOBILIZATION

Spinal immobilization should be performed on the basis of mechanism of injury and the patient's symptoms.

- If the patient has major mechanism of injury (MOI) or MOI cannot be ruled out, then always immobilize.

- Is the patient significantly obtunded?
  - Head Injury.
  - Drugs or Alcohol
- **Does the patient have neck / midline spine tenderness?**
- Does the patient have a neurologic deficit?

YES → Evaluate the **M**echanism **o**f **I**njury.

NO → No immobilization necessary.

Evaluate the Mechanism of Injury →

High-Risk of Injury. (defined below) → Provide **Full Spinal Immobilization.**

Low-Risk of Injury. (defined below) → Apply C-Collar, transport in position of comfort.

When in doubt, immobilize.

Transport as indicated.

Examples of High-Risk Mechanism of Injury:

- High-speed MVC.
- Rollover MVC.
- Intrusion into patient compartment.
- Death in the vehicle.
- Fall 2x patient height.
- Fall onto head.
- Auto vs. Pedestrian.
- Diving Injury.
- Penetrating trauma *with* motor/sensation deficits.

Examples of Low-Risk Mechanism of Injury:

- Low speed MVC.
- Fall from standing position.
- Penetrating trauma *without* motor/sensation deficits.
- Ambulatory at the scene.

Pediatric Patients Low Risk Injury:

No board or collar if:

- Fall from standing.
- Fall from bed.
- MVA in car seat.
- Neurologically normal for age.
- No apparent serious injury.

SPECIAL CONSIDERATION IN PREGNANT PATIENTS:

If spinal immobilization is required, attempt to tilt the patient 15 – 30 degrees to her left side in order to keep the baby in vitro from compressing the patient's inferior vena cava, causing hypotension.

If the patient refuses immobilization, all risks are to be explained to the patient and documented in the narrative along with a witness' signatures.

Any deviation from this protocol requires contact with Online-Line Medical Control.

TREATING CHILDREN FROM METH HOMES

TREATING CHILDREN
FROM HOMES WITH METH LABS

There are two main concerns when treating children taken from homes with meth labs - **medical evaluation** and **decontamination.**

MEDICAL EVALUATION

Determining the need for immediate evaluation by a health care provider can be accomplished by assessing whether the child is **asymptomatic** or **symptomatic**.

Asymptomatic

Asymptomatic children do not need to be immediately evaluated by a physician.
If the paramedic or police officer finds a child who is completely asymptomatic, there is no need to take that child to the emergency department.

In cases where medical evaluation is required before the child can be placed in an appropriate social setting, then the emergency department evaluation requires only a history and physical exam.

Subsequent to removal from the home, children can be seen at their pediatrician's office, where issues like neglect may also be addressed.

Symptomatic

For the symptomatic child, medical evaluation should happen immediately and should include a physical exam and history. All laboratory testing should be based on findings from the physical exam and/or other concerning symptoms.

Although there are concerns about the long-term effects of these exposures, there is not an initial evaluation or intervention that can identify or change the long-term outcome.

DECONTAMINATION

If there is residue on the clothes of an asymptomatic child, clothing can be changed prior to transfer. Alternately, the children can be given something disposable to sit on (newspaper, diaper, etc.) and clothes can be changed post-transfer. Regardless of symptom status, children should take a shower at some point following removal, but do not need to be decontaminated at the scene.

TIPS FOR TREATING AND DEALING WITH CHILDREN TAKEN FROM HOMES WITH METH LABS:

- Always take the least traumatic approach.
- Do not take toys or any materials from the house.

Poisons commonly found at meth production sites:

- Solvents
- Ephedrine
- Acids
- Iodine
- Lye
- Phosphorus

The dangers from living in a home that houses a methamphetamine lab can include:

- Injury or death from fire or explosions caused by flammable materials used in the lab.
- Chemical burns
- Poisoning from ingestion or absorption of chemicals
- Upper respiratory symptoms, headaches and rash from exposure to chemicals in the environment
- Long-term effects, such as asthma and neurological problems
- Malnutrition
- Emotional and/or physical neglect

A medical toxicologist at Tennessee Poison Center is always available to answer questions. Call 1-800-222-1222 for help 24/7.
Tennessee Poison Center is a program of Vanderbilt University Medical Center.

For more information on methamphetamine, please visit
www.TNPoisonCenter.org or www.MethFreeTN.org

OB / GYN COMPLAINTS (NON – DELIVERY OR GYNECOLOGICAL ONLY)

- Patient Para (number of live births) and Gravida (number of pregnancies)?
- Term of pregnancy in weeks, EDC, Multiple births expected or history?
- Vaginal bleeding (how long and approximate amount)?
- Possible miscarriage/products of conception?
- Pre-natal medications, problems, and care?
- Last menstrual cycle?
- Any trauma prior to onset?
- Lower extremity edema?

1. Patient positioning appropriate for condition.
2. Oxygen and airway maintenance appropriate to the patient's condition.
3. Control hemorrhage as appropriate.
4. **IV Lactated Ringers (LR) TKO unless signs of shock, then 20 mL/kg fluid bolus of LR, consider Glucose check.**

Abruptio Placenta:

- Multiparity
- Maternal hypertension
- Trauma
- Drug use
- Increased maternal age
- History
- Vaginal bleeding with no increase in pain
- No bleeding with low abdominal pain

1. Position Patient in the left lateral recumbent position.
2. **Pregnant patients in 2nd and 3rd trimesters with blunt trauma (MVA-seatbelt), should never refuse transport, as they are at risk of abruption placenta.**

Placenta Previa:

- Painless bleeding which may occur as spotting or recurrent hemorrhage
- Bright red vaginal bleeding usually after 7th month
- History
- Multiparity
- Increased maternal age
- Recent sexual intercourse or vaginal exam
- Patient para (number of live births) and gravida (number of pregnancies)
- Term of pregnancy in weeks
- Pre-natal medications, problems, and care
- History of bed rest
- Placenta protruding through the vagina

1. Oxygen and airway maintenance appropriate to the patient's condition.
2. Position of comfort.

NOTE:

Any painless bleeding in the last trimester should be considered Placenta Previa until proven otherwise. If there are signs of eminent delivery membrane rupture is indicated followed by delivery of the baby. The diagnosis of eminent delivery depends on the visual presence of the baby's body part through the membrane.

OBSTETRIC EMERGENCIES – ACTIVE / IMMINENT DELIVERY

- Assess ABC's, stabilize as needed.
- Pulse oximetry.
- Cardiac monitor.

Apply oxygen 100% via NRB.

Begin transport without delay, using caution to not risk the safety of EMS personnel and patients while in transport.

Notify the receiving facility ASAP
Ask pertinent history, as defined below.

Obtain IV access – at least a 20 ga. Between wrist and AC is preferred, LR bolus as needed to maintain normal BP.

Perineal examination (DO NOT perform an internal vaginal exam):
- If in active labor with no bleeding or crowning, transport as indicated.
- If vaginal bleeding and/or signs of shock, transport emergency.

If delivery is imminent:
(Contractions q 3-5 minutes, lasting 30-60 sec)
- Prepare area for delivery (OB kit)
- Keep ambulance as warm as possible
- Prepare mother for delivery, preserve dignity.

Use gentle pressure to control delivery.

- Prevent an explosive delivery.

When head delivers, suction airway:

- Using bulb syringe, suction mouth, then nose!
- Suction any meconium from the airway ASAP!

*****Keep the baby level with the mother*****

Following delivery of the baby, ensure to:
- **Dry vigorously to stimulate breaths.**
- Maintain airway.
- Protect baby from fall risk.
- Protect from risk of hypothermia.
- Wrap baby in blanket to keep warm.
- Apply a head cover to baby to preserve body heat.
- Waiting 1-2 minutes before clamping the umbilical cord.
- **Clamp the umbilical cord at 8 and at 10 inches from the baby and cut between clamps once.**
- Consider allowing baby to nurse if mother is willing, and if there is no history of drug use.
- **Consult on-line medical control as needed.**

Check **APGAR** at 1 and 5 minutes post-delivery:
- REFERENCE on next page.

Allow placenta to deliver:
(May take up to 20 minutes normally)
- **Massage uterine fundus** (lower abdomen)
- Observe and treat signs of shock with increased deliver of oxygen and IV fluids.

- Reassess for postpartum hemorrhage.
- Reassess cord for bleeding, if bleeding add additional clamp and continue to monitor.
- Check glucose in baby's heel, **not finger**.
  - Normal neonate glucose level is >50.
- **Refer to the Neonatal Resuscitation Protocol if needed.**

Transport as indicated.

Pertinent history – gather SAMPLE, and:

- History of problems with pregnancy.
- Last menstrual period and due date.
- Number of pregnancies and deliveries.

OBSTETRIC EMERGENCIES – ACTIVE / IMMINENT DELIVERY (continued)

Considerations:

1. The greatest risks to the newborn infant are airway obstruction and hypothermia. Keep the infant warm (silver swaddler or space blanket), dry, covered, and the infant's airway maintained with bulb syringe. Always remember to squeeze the bulb prior to insertion into the infant's mouth or nose.
2. The greatest risk to the mother is post-partum hemorrhage. Watch closely for signs of hypovolemic shock and excessive vaginal bleeding.
3. Spontaneous or induced abortions may result in copious vaginal bleeding. Reassure the mother, elevate legs, treat for shock, and transport.
4. Record a blood pressure and the presence or absence of edema in every pregnant woman you examine, regardless of chief complaint.

*****Complete patient care reports on BOTH mother and child.*****

OB EMERGENCIES – ABNORMAL / COMPLICATED DELIVERY

Please give report to the receiving facility as soon as possible.

Consult with On-line Medical Control as needed – High Risk OB patients, or those with complications may need to be transport to facilities that can provide specialty care. (*Vanderbilt, Centennial Women's, St. Thomas Mid-Town)*

Amniotic Sac Presentation:

1. Place patient in a position of comfort.
2. Amniotic sac –
   - if no fetus or head of fetus is visible, cover presenting sac with moist, sterile dressing.
3. Contact on-line medical control ASAP.

Nuchal Cord (umbilical cord around the neck):

1. Carefully remove cord from around baby's neck by slipping cord over the baby's head if possible.
2. Attempt to prevent creating a knot in the cord.
3. Prevent cord from strangulating the neonate during delivery.

Breech Presentation (Buttocks or body, not head coming out):

1. Place patient in best possible position and **transport EMERGENCY**.
2. Allow the delivery to progress spontaneously – DO NOT PULL!
3. Support the infant's body as it delivers.
4. If the head delivers spontaneously, deliver the infant as noted in 'Normal Delivery'.
5. If the head does not deliver within 3 minutes, insert a gloved hand into the vagina an airway for the infant.
6. DO NOT remove your hand until relieved by a Higher Medical Authority.
7. Contact on-line medical control ASAP.

Limb Presentation:

1. Position the mother in a supine position with the head lowered and pelvis elevated, **transport EMERGENCY**.

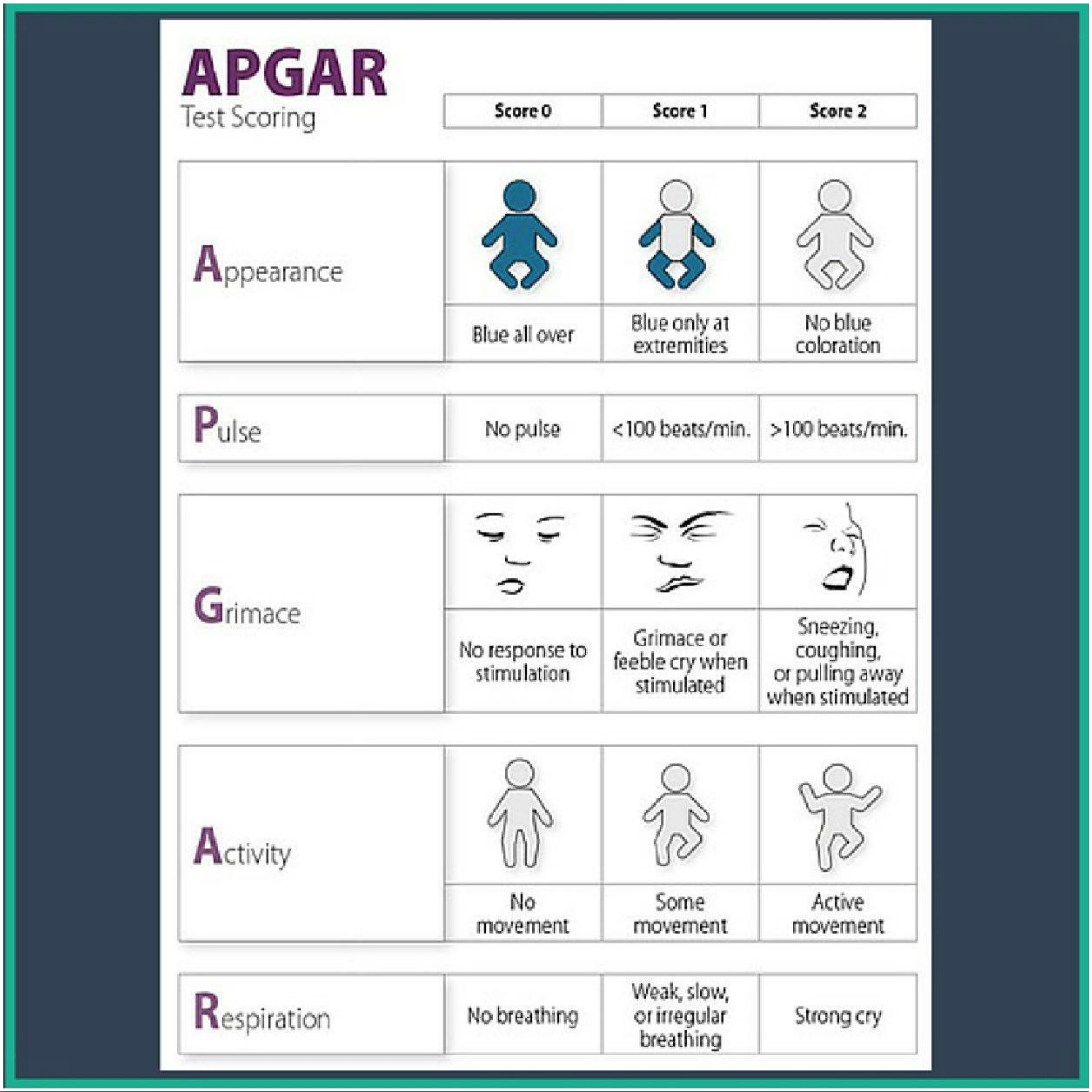
Meconium Staining:

1. Do not stimulate respiratory effort before suctioning the oropharynx.
2. Suction the **mouth then the nose** **(using a meconium aspirator)** while simultaneously providing Oxygen 100% by blow by method and while maintaining the airway appropriate to the patient's condition.
3. Obtain and APGAR score after airway treatment priorities. Score one minute after delivery and at five minutes after delivery. (Time permitting)

Prolapsed Cord:

1. Position the mother with hips elevated.
   - Knees to chest.
   - Transport in a supine position with her hips elevated as much as possible on pillows.
2. Instruct mother to pant with each contraction, which will prevent her from bearing down.
3. Check for a pulse in the cord.
   - **If no pulse** – insert a gloved hand into the vagina and gently push the infant's head off the cord. While pressure is maintained on the head cover the exposed cord with a sterile dressing moistened in saline. Transport immediately and DO NOT remove your hand until relieved by hospital staff.
   - **If pulse present** – cover exposed cord with moist dressing.
4. Contact on-line medical control ASAP if time and patient condition allows.

APGAR Score Reference Chart



APGAR
Test Scoring

	Score 0	Score 1	Score 2
Appearance	Blue all over	Blue only at extremities	No blue coloration
Pulse	No pulse	<100 beats/min.	>100 beats/min.
Grimace	No response to stimulation	Grimace or feeble cry when stimulated	Sneezing, coughing, or pulling away when stimulated
Activity	No movement	Some movement	Active movement
Respiration	No breathing	Weak, slow, or irregular breathing	Strong cry

Score of 0-3 = Severely depressed, RESUSCITATE!!!

Score of 4-7 = Moderately depressed, STABILIZE

ASSESS A.P.G.A.R. SCORES AT 1 MINUTE AND 5 MINUTES POST PARTUM!

NEONATAL (NEWBORN) RESUSCITATION

This applies to term and pre-term newborn patients who fail to respond to initial stimulation and are in need of stabilization or resuscitation efforts. The standing order here applies to neonatal patients in general, which shall include patients who are less than 1 month of age.

Within the first 30 seconds:

- As soon as the baby is born: vigorously dry the infant and provide warmth (heat on high, cover with blanket).
- Position the infant to open the airway (sniffing position, careful not to hyperextend the neck).
- Clamp and cut the cord per OB/delivery protocol.
- If excessive secretions AND signs of compromise are present, clear with airway with a bulb syringe.
  - Routine suctioning of the oropharynx and nasal pharynx is no longer recommended.
  - If meconium staining is present AND the newborn is not vigorous (weak or absent respiratory efforts, weak or absent muscle tone, heart rate less than 100/min), tracheal suctioning may be considered.
- Stimulate breathing (flicking the soles of the baby's feet or rubbing the baby's back).

↓

Assess respirations:

- If inadequate or gasping respirations are present, assist ventilations at a rate of 40 to 60 breaths per minute using an appropriately sized BVM attached to high flow O2. (Careful to not over-inflate).
- If neonate is premature, titrate O2 flow, initiate on room air, do not exceed 65% FiO2 (6 lpm)
- If the respirations are shallow and slow, attempt a 1-minute period of stimulation while administering oxygen via blow-by method.
  - If respirations do not increase, initiate positive pressure ventilations as directed above.

↓

Assess heart rate:

- If heart rate is less than 60 beats per minute, begin chest compressions, ventilate with 100% FiO2.
  - Compression-to-ventilation ratio of 3:1 in neonatal resuscitation, compress at 120/min.
  - Compressions should be discontinued when heart rate is higher than 60 beats/min (with pulse).

↓

Advanced Resuscitation:

- ❖ Consider advanced airway (one attempt only) for:
  - ❖ Persistent apnea
  - ❖ Central cyanosis
  - ❖ Bradycardia (HR < 100)
- ❖ If HR persistently < 60:
  - ❖ Continue CPR.
  - ❖ Initiate IV/IO normal saline.
  - ❖ Administer 1:10,000 Epinephrine 0.02 mg/kg (0.2 mL/kg) IV/IO every 3-5 minutes as needed.
- ❖ Obtain blood glucose level (perform heel stick, <u>not finger</u>), if < 50, administer Dextrose 10% 2-4 mL/kg IV/IO.
- ❖ Consider IV isotonic crystalloid fluid (NS, LR, D5W, etc.) administration at 10 mL/kg as needed.

PRE-ECLAMPSIA / ECLAMPSIA

- Assess ABC's, stabilize as needed.
- Pulse Oximetry.

Signs and Symptoms:

- Headache.
- Double-vision or seeing spots.
- Blood Pressure greater than 140/90.
- Generalized swelling of the face, arms, and legs.

Administer oxygen and begin external warming?

Monitor cardiac rhythm:

- Treat life-threatening arrhythmias.

Obtain IV access, consider 2 lines as patient's condition presents.

Try to have the patient relax as much as possible.

Is systolic BP > 140 and/or diastolic BP > 90?

YES

Is the patient > 20 weeks pregnant OR < 2 weeks post-delivery without a history of seizures?

YES

Mix 4 grams of Magnesium Sulfate in a 100 mL bag of NS and infuse IV/IO over 20 minutes.
This can be given in conjunction with Versed and Labetalol, if the patient becomes eclamptic and starts having seizures.

NO

Administer **Labetalol** 10 mg slow IV.

May repeat at 20 mg slow IV every 10 minutes as needed.

NO

Transport as indicated.

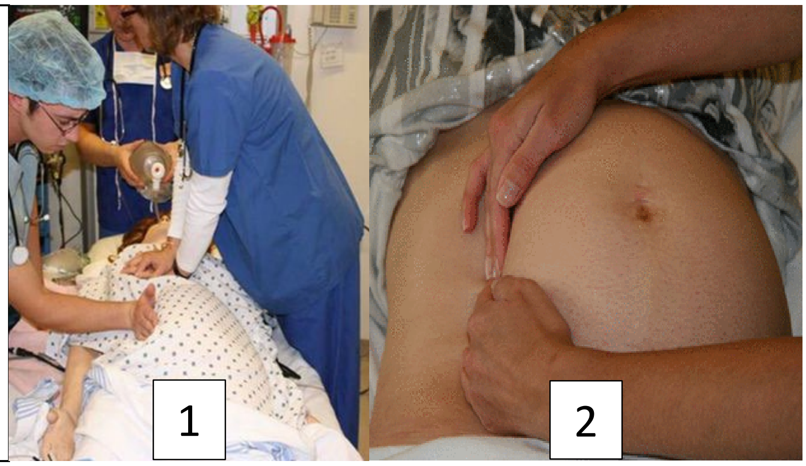
MATERNAL ACLS / CARDIAC ARREST IN PREGNANCY

Management of the Unstable Pregnant Patient:
- The patient should be placed in a full left lateral decubitus position to relieve aortocaval compression.
- Intravenous access should be established above the diaphragm to ensure that the intravenously administered therapy is not obstructed by the gravid uterus.
- Precipitating factors should be investigated and treated.
- Pregnant patients are at an **increased risk of aspiration**, hypoxemia and pulmonary edema.
- Systemic hypotension can overwhelm compensatory mechanisms, which attempt to maintain uterine blood flow.

"Manual LUD"

Left
Uterine
Displacement

1. One-hand technique
2. Two-hand technique



Chest Compressions in Pregnancy:
- Guidelines for hand placement, rates, and ratios are the same as with non-pregnant adult patients.
- Keep supine, implement manual LUD during compressions, starting at ≈12 to 14 weeks of gestational age.
- There is no literature examining the use of mechanical CPR devices in pregnancy, this isn't advised at this time.

Transport / Operations:
- Transport to a facility that can perform a cesarean delivery may be required when indicated (For out-of-hospital cardiac arrest or cardiac arrest that occurs in a hospital not capable of cesarean delivery).
- If available, transport should be directed toward a facility that can perform PMCD, but transport should not be prolonged by >10 minutes to reach an ER with more capabilities. (PMCD = perimortem cesarean delivery).

Electrical Therapy (Defibrillation) Issues During Pregnancy:
- Current defibrillation protocols should be used in the pregnant patient as in the non-pregnant patient.
- The lateral defibrillator pad/paddle should be placed under the breast tissue when possible.

Breathing and Airway Management in Pregnancy:
- Oxygen reserves are lower and metabolic demands are higher; earlier ventilatory support may be necessary.
- Intubation starting with an ETT with a 6.0- to 7.0-mm inner diameter is recommended.
- Upper airway edema and friability (weakening of tissue) occur as a result of hormonal effects and may reduce visualization during laryngoscopy and increase the risk of bleeding.
- Renal blood flow increases, stressing the kidneys, increasing the tendency for pulmonary edema to develop.

Emergency (ACLS) Medications in Pregnancy:
- In the setting of cardiac arrest, no medication should be withheld because of concerns about fetal teratogenicity.
- Drug metabolism is altered in pregnancy. However, there are no current recommendations for altering dosages of routine ACLS drugs. **Use standard drugs and doses when indicated.**